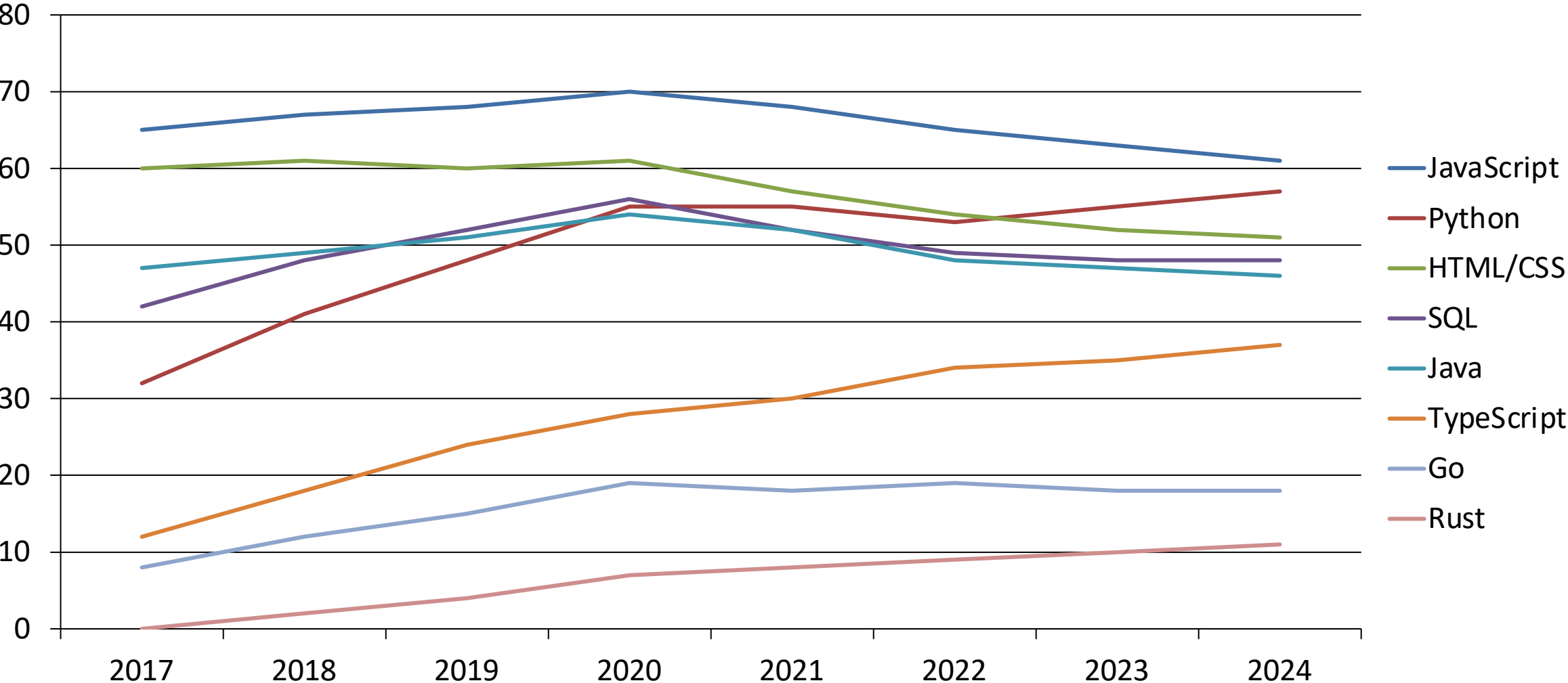


State of Developer Ecosystem 2024

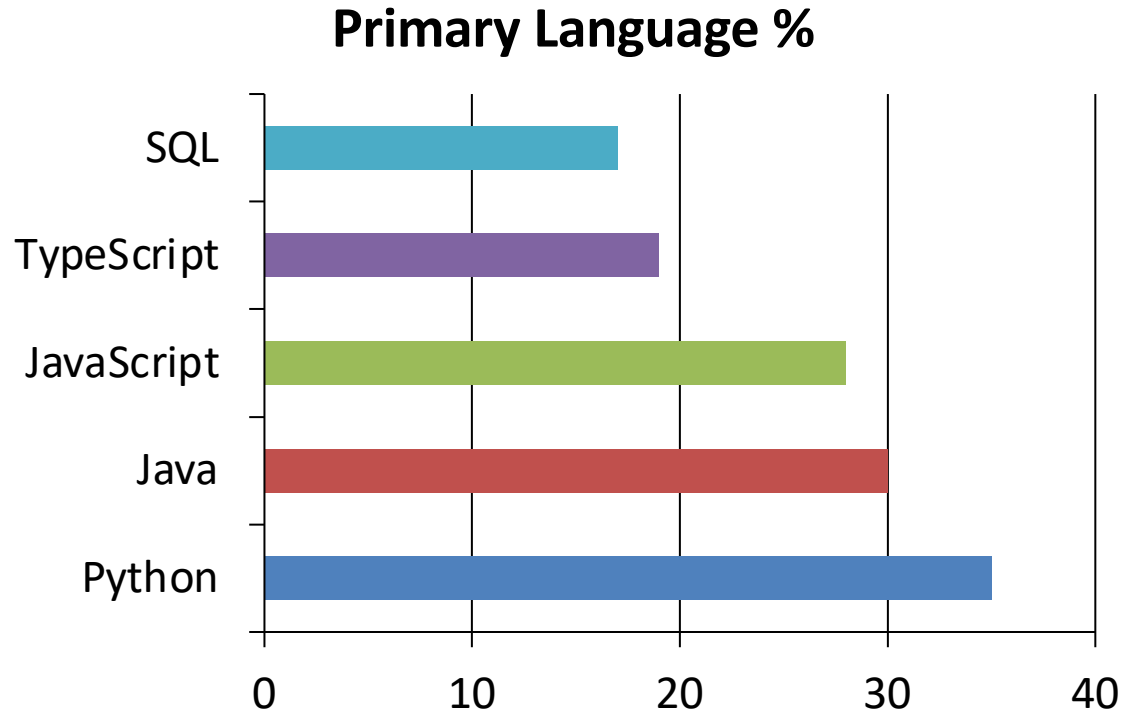
Language Trends, Satisfaction, and Strategic Shifts

Based on JetBrains survey of 23,262 developers worldwide

Python Closes the Gap: 8 Years of Language Usage Shifts



Python Is Now the #1 Primary Language at 35%



Ecosystem Lock-In Effect

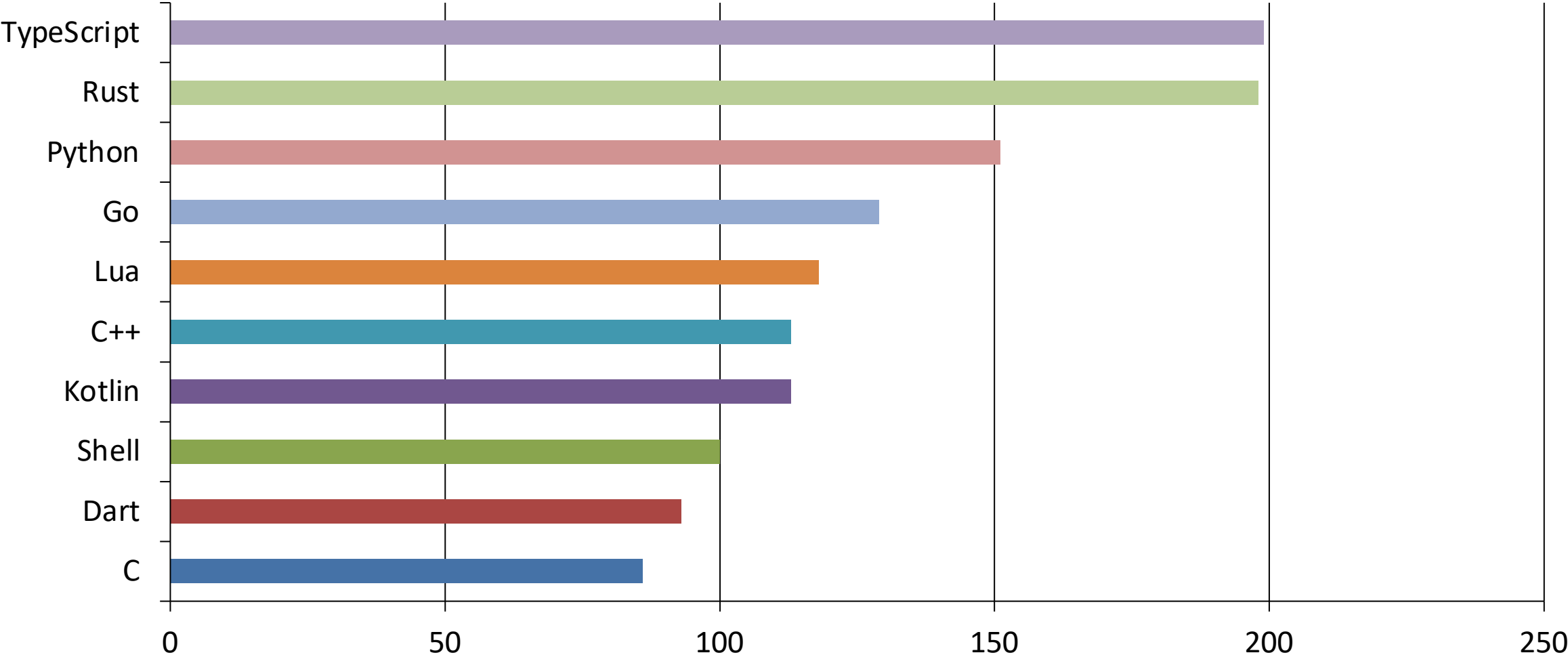
Adoption Intent:

Python: 6% planning to adopt
TypeScript: 5% planning to adopt
Java: 4% planning to adopt
SQL: 4% planning to adopt

Despite Python's rise, developers rarely plan to switch their primary stack — ecosystem lock-in keeps teams stable.

TypeScript and Rust Top Developer Satisfaction

Satisfaction Score



Each Language Owns a Domain

Domain	Leading Language	Market Share
AI / Machine Learning	Python	31%
Data Processing & Analytics	Python	17%
Data Science / Analytics	Python	28%
User Interface	TypeScript	62%
APIs & Services	Java	49%
Application Logic & Workflows	Java	55%
System Tools & Components	C++	27%
Integrating with APIs & Services	Python	38%

What This Means for Your Technology Strategy in 2025

1. Invest in Python for AI/ML teams

Python is the undisputed leader in AI/ML (31%) and data science — hire and develop accordingly.

2. TypeScript is the future of frontend and full-stack

At 62% for UI development and high satisfaction scores, TypeScript skills are critical.

3. Rust and Go are the growth plays for infrastructure

Both languages show rising usage and satisfaction — ideal for cloud-native and systems work.

4. PHP and Ruby hiring pools are shrinking

Long-term decline suggests reduced talent availability — plan migration paths if on these stacks.

5. Developer satisfaction correlates with type safety

TypeScript (199) and Rust (198) satisfaction scores show type-safe languages retain talent better.